INCH-POUND

MIL-I-15864K(NU)
7 September 1990
SUPERSEDING
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18 January 1984

MILITARY SPECIFICATION

INSIGNIA, EMBROIDERED (MEN AND WOMEN)

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirement for Schiffli and hand embroidered sleeve insignia worn by enlisted men and women of the Navy and Navy personnel attached to the Marine Corps units.

1.2 Classification. Embroidered insignia shall be of the following types and classes. (see 6.2):

Type I - Badge, Rating
  Class 1 - Chief Petty Officer
  Class 2 - Petty Officer - First Class
  Class 5 - Master Chief Petty Officer
  Class 6 - Senior Chief Petty Officer
  Class 7 - Master Chief Petty Officer of the Navy

Type II - Specialty Mark and Striker Mark (see Table III)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8455

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Type V - Group - Rate Mark
  Class 2 - Two Stripes
  Class 3 - Three Stripes

Type VI - Stripe, Service
  Class A - 7 inches long
  Class B - 5-1/4 inches long
* Class C - 7 inches long (synthetic metallic yarn)
* Class D - 5-1/4 inches long (synthetic metallic yarn)

Type VIII - Insignia, Grade

Type X - Badge, Rating, Utility, Petty Officers, First Class

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
PPP-B-676	- Boxes, Set-up

MILITARY

MIL-C-823	- Cloth, Serge, Wool, Wool and Nylon, Polyester and Wool
MIL-C-3738	- Cloth Elastique, Wool
MIL-C-10176	- Cloth, Gabardine, Wool, Polyester, and Wool
MIL-C-16290	- Cloth, Melton, Wool (Moth Proofed)
MIL-L-17507	- Lace, Ornamental
MIL-E-17568	- Embroidery Materials, Metallic and Synthetic Metallic
MIL-B-17757	- Box, Shipping, Fiberboard (Modular Sizes)
MIL-I-18496	- Insignia, Embroidered (Women's)
MIL-C-21115	- Cloth, Tropical, Wool, Polyester/Wool
MIL-C-21881	- Cloth, Poplin, Polyester and Cotton
* MIL-C-24910	- Cloth, Tropical: Polyester and Wool
MIL-C-29127	- Cloth, Twill, Polyester/Cotton (Crease-Resistant Finish)
MIL-C-43482	- Cloth, Poplin: Cotton and Polyester, Water Repellent
MIL-C-87052	- Cloth, Twill, Polyester, White

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STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-129	- Marking For Shipment and Storage
* MIL-STD-2073-1	- DoD Materials Procedures Development and Application of Packaging Requirements
* MIL-STD-2073-2	- Packaging Requirement Codes

* (Unless otherwise indicated, copies of Federal and Military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

(Stitch enlargement drawing numbers are as indicated in Table III. Figures 1 through 16 are illustrations of Navy Clothing and Textile Research Facility stitch enlargement drawings, and are attached for information only.)

* U.S. POSTAL MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

* 2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808.)

THE COLOR ASSOCIATION OF THE UNITED STATES

Standard Shades for Heraldic Yarns for Embroidery and Ribbons
Standard Shades for Sewing Threads

(Applications for copies should be addressed to the Color Association of the United States, 343 Lexington Ave., New York, NY 10016-0927.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Materials.

3.3.1 Cloth, Wool, Melton. The melton shall conform to the requirements for type I of MIL-C-16290, except the color shall match Blue 3346.

* 3.3.2 Cloth, Wool, Elastique. The elastique shall conform to the requirements for type I or type II, class 2 of MIL-C-3738. The color shall match Green 2222.

* 3.3.3 Cloth, Gabardine, Wool, Polyester and Wool. The gabardine shall conform to the requirements for type I, class 5 of MIL-C-10176. The color shall match Blue 2316(MC). The fabric for the Marine Corps shall conform to type II, class 12 of MIL-C-10176. The shade shall be Green 2212.

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* 3.3.4 Cloth, Tropical. The cloth, tropical shall conform to the requirements of type III, class I, Khaki 3729 or type III, class 3, Blue 3346 of MIL-C-21115. The color for the Marine Corps shall be Green 2241.

3.3.5 Cloth, Wool, Serge. The serge used shall conform to the requirements of type I, class 5, of MIL-C-823. The colors shall match Green 2234 (MC) and Blue 3346 (Navy).

3.3.6 Cloth, Twill, Polyester/Cotton. The twill material shall conform to the requirements of class 1 or 2 of MIL-C-29127. The color for class 1 shall be Khaki 2101 (MC). The color for class 2 shall be White 3053.

3.3.7 Cloth, Poplin, Polyester, and Cotton. The material shall conform to the requirements of type II, class 2 of MIL-C-21881. The color shall match White 3013.

3.3.8 Cloth, Poplin, Cotton and Polyester, Water Repellent. The poplin material shall conform to the requirements of MIL-C-43482, except that the requirements for water repellency and seam efficiency shall not apply. The color shall match Blue 3329.

* 3.3.9 Cloth, Tropical: Polyester and Wool. The cloth to be used as a background for embroidered designs shall conform to the requirements of MIL-C-24910. The color shall be Blue 3346.

NOTE: Background fabrics applicable to women's striker marks shall be specified in MIL-I-18496, Insignia, Embroidered (Women's)

* 3.3.10 Cloth, Twill, Polyester. The cloth to be used as a background for embroidered designs shall conform to the requirements of MIL-C-87052. The color shall be White 3053.

NOTE: Background fabrics applicable to women's striker marks shall be specified in MIL-I-18496, Insignia, Embroidered (Women's)

3.3.11 Embroidery materials, metallic and synthetic metallic. The embroidery materials shall conform to the requirements for type I, II, or III of MIL-E-17568.

3.3.12 Lace, ornamental. The gold lace shall conform to the requirements for type II, style A, class 2, 3, 4, or 5, 3/8 inch in width, one or two vellum of MIL-L-17507. The color shall match Gold 3903.

3.3.13 Buckram. The cotton buckram shall be natural or bleached white, plain weave and shall conform to the following requirements when tested as specified in 4.4.1.

Type A - - - 50 (±3) ends and 50 (±3) picks per inch. The weight shall be 4.8 (±1.0) ounces per square yard.
Type B - - - 20 (±2) ends and 15 (±2) picks per inch. The weight shall be 6.75 (±1.0) ounces per square yard.

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3.3.13.1 Buckram backing. Unless otherwise specified, buckram backing shall be used to reinforce the following areas of insignia:

Type A - Hand embroidered bullion or synthetic metallic stars, eagle and specialty mark of rating badge.
Sewn on type gold lace chevrons and service stripes.

* Type B - One layer of bleached buckram when the white polyester background cloth conforming to MIL-C-87052 is used.

* Types A and B - Buckram reinforcement is not required but may be used on Schiffli embroidered synthetic metallic stars, eagle, specialty mark, chevrons, and service stripes.

3.3.14 Yarns - Schiffli embroidered insignia.

* 3.3.14.1 Rayon yarn. The rayon yarn shall be bright, continuous filament, regenerated cellulose type, 75/3 (± 5) denier Schiffli yarn. The rayon yarn for the eagle tongue for Petty Officer 3rd Class and above shall be bright, continuous filament, regenerated cellulose type, 100/2 (± 5) denier or 75/3 (± 5) denier Schiffli yarn. Testing shall be as specified in 4.4.1.

* 3.3.14.1.1 Colors. Cable numbers for the colored yarns shall be as follows (see 2.2 and 6.2). The use of sulfur dyes is prohibited.

White	- Cable No. 67101
Golden Yellow	- Cable No. 67104
Scarlet	- Cable No. 67111
Blue Bird	- Cable No. 67117
Emerald Green	- Cable No. 67128
Dark Blue	- Cable No. 67126

* 3.3.14.1.1.1 Colorfastness. The yarn shall show fastness to light, laundering, wet-dry cleaning, bleaching and dry and wet heat (hot pressing), equal to or better than the standard sample when tested as specified in 4.4.4. As a limit of acceptability or when no standard sample is available, the yarn shall show good fastness to light, laundering, wet-dry cleaning, bleaching and dry and wet heat (hot pressing) when tested as specified in 4.4.1.

* 3.3.14.1.2 Matching. The dyed yarns shall match the available standard shade sample when viewed under filtered tungsten lamps that approximate artificial daylight, having a correlated color temperature of 7500 (± 200) K with illumination of 100 (± 20) foot candles and shall be a good match to the standard shade sample under incandescent lamplight at 2300 (± 200) K.

* 3.3.14.2 Bobbin thread. The bobbin thread used in Schiffli embroidery shall either be a good commercial grade of two-ply (minimum) undyed cotton thread or continuous filament undyed (white) nylon thread equal to or better than the quality of the specified cotton thread when tested as specified in 4.4.1.

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* 3.3.14.3 Synthetic metallic yarn. As an alternate to the yarn specified in 3.3.14.1, the yarn for the type VI Service Stripe, Classes C and D (see Table III) shall conform to type III, Class 8 or 11 of MIL-E-17568. Testing shall be as specified in MIL-E-17568. Design and construction requirements shall be as specified in 3.6.5.

* 3.3.15 Thread, cotton. The white or colored thread used for cover stitching and hemming shall conform to the requirements of V-T-276 and as specified in Table I. Unless otherwise specified, the color of the thread shall match the color of the fabric used as the background cloth.

* 3.3.15.1 Colorfastness. The thread shall show fastness to dry and wet heat (hot pressing), equal to or better than the standard sample when tested as specified in 4.4.1. As a limit of acceptability or when no standard sample is available, the yarn shall show "good" fastness to dry and wet heat (hot pressing) when tested as specified in 4.4.1.

Table I - Threads

Type	Ply	Ticket Number		Operations
		Needle	Lower	
1A1	3	40	-	Attaching metallic embroidery materials by hand
1C1	3	00	00	Hemming and cover stitching (white)
1C2	3	00	00	Hemming, attaching gold lace, coverstitching (colors)

3.3.16 Form, fiberboard. When needed, the form used to raise the design of hand embroidered metallic insignia shall be a commercial grade fiberboard of suitable thickness and shall be free of sulfur (see 4.4.1). The face side of the board shall be a color to approximately match the color of the embroidery material covering it.

3.3.17 Stiffener. When needed, hand embroidered insignia may be treated on the reverse side with a stiffener to facilitate processing or to prevent raveling of the yarn. The stiffener shall not affect the color of the face of the insignia or the background fabric while in storage.

3.3.18 Stitches, seams, and stitchings. Stitch types shall conform to FED-STD-751. Stitch type 301 used for hemming shall be 1/8 (± 1/32) inch from the edge. Stitch type 304 used for cover stitching shall be 1/8 (± 1/32) inch wide. Stitches per inch shall be 12 to 14 for general stitching and 45 to 48 for cover stitching.

* 3.4 Description. The insignia specified in this specification shall be either Schiffli embroidered with rayon yarn and/or synthetic metallic material in accordance with the applicable Government stitch enlargements (see 3.5.3), or hand embroidered with metallic (bullion) or synthetic metallic embroidery materials in accordance with approved guide samples (see 3.1).

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* 3.4.1 Schiffli construction.

3.4.1.1 Ground and overlay stitches. The ground and overlay stitches shall be executed in accordance with the applicable stitch enlargement (see 3.5.3), and shall completely cover the prescribed background design area in a closely, densely, and tightly drawn manner to produce a clear sharp execution of the design with none of the background cloth or underlay stitches visible.

3.4.1.1.1 Chevron (rating badge), marks (group rate), and stripe (service). Stitch execution of chevron marks and stripes shall be of the ground embroidery type. The number of visible rows shall be as indicated in the applicable stitch enlargement (see 3.5.3).

3.4.1.2 Underlay stitches. Underlay stitches, where required, shall be inserted in the direction and manner most suitable for proper design execution and shall not be less than the number indicated on the applicable stitch enlargement. The raised areas of the design where underlay is required, shall be equal to the guide sample (see 3.1) in fullness and depth.

3.4.1.3 Stiele border stitches. When required, Stiele border stitches shall be executed in accordance with the applicable stitch enlargement or guide sample (see 3.1 and 3.5.3) and shall be tightly drawn and well executed to produce a clear, sharp border and shall completely cover the background embroidery and cloth.

3.4.1.3.1 Chevron (rating badge), marks (group rate) and stripes (service). The edges of each chevron, mark and stripe shall be finished with a 1/16 inch wide Stiele border stitch.

3.4.2 Construction with metallic (bullion) and synthetic metallic embroidery materials. When the eagle, stars and specialty marks are constructed with metallic (bullion) and synthetic metallic embroidery materials the designs shall be produced using component materials specified in 3.3.11, 3.3.13.1, 3.3.16, 3.3.17 and Table II.

3.4.3 Gold lace chevrons and service stripes. Insignia utilizing gold lace shall be produced by combining the background cloth with type A buckram (see 3.3.13). The gold lace shall be superimposed over the background both in the prescribed manner (see 3.1) and stitched along the sides with thread of color to match the gold lace and using stitch type 301. The ends of the gold lace shall fit into slits cut into the background cloth and the entire length of slits completely coverstitched with stitch type 304 and thread matching the shade of the background material.

3.4.4 Finishing.

3.4.4.1 Hemmed edges. Unless otherwise specified, all insignia shall have all outer edges of background cloth hemmed with a 3/8 to 1/2 inch turnunder, using stitch type 301 and seam type EFa-1. The color of thread shall match the background cloth.

3.4.4.2 Unhemmed edges. Insignia produced on melton fabric shall be cut to size, no hemming is required.

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3.4.5 Spacing. All insignia covered by this specification shall be centered on the background.

3.5 Color combination and design.

3.5.1 Color combination. The color combination of insignia shall be in accordance with Table II as indicated by the color combination code letter.

Table II - Color and Material for Design

Color Combination Code Letter	Background Cloth	Rating Badge				Type of Design		
		Eagle	Specialty Mark	Chevrons	Stars	Striker Mark Specialty	Group Rate	Service Stripe
* A	Blue Serge	Silver Bullion or Syn Met	Silver Bullion or Syn Met	Gold Syn Lace or Schiffli Emb	Silver Bullion or Syn Met	N/A	N/A	Gold Syn Lace or Schiffli Emb
* A (MCPO)	Blue Serge	Silver Bullion or Syn Met	Gold Bullion or Syn Met	Gold Syn Lace or Schiffli Emb	Gold Bullion or Syn Met	N/A	N/A	Gold Syn Lace or Schiffli Emb
B	Blue Serge	Silver Bullion or Syn Met	Silver Bullion or Syn Met	Scarlet	Silver Bullion or Syn Met	N/A	N/A	N/A
C	Blue Serge	N/A	N/A	N/A	N/A	N/A	N/A	Scarlet
D	Bleached Twill	Blue	Blue	Blue	Blue	Blue	Blue	Blue
DD	Bleached Poplin	Blue	Blue	Blue	Blue	Blue	Blue	N/A
* DDD	White Polyester Twill	Blue	Blue	Blue	Blue	Blue	Blue	N/A
E	Khaki Twill	Blue	Blue	Blue	Blue	Blue	Blue	Blue
F	Khaki Tropical	Silver Bullion or Syn Met	Silver Bullion or Syn Met	Blue	Silver Bullion or Syn Met	N/A	N/A	Blue

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Table II - Color and Material for Design (cont'd)

Color Combination Code Letter	Background Cloth	Rating Badge				Type of Design		
		Eagle	Specialty Mark	Chevrons	Stars	Striker Mark Specialty	Group Rate	Service Stripe
G	Green Elastique	Blue	Blue	Blue	Blue	N/A	N/A	Blue
H	Blue Melton	White	White	Scarlet	N/A	White	White	N/A
HH	Blue Serge	White	White	Scarlet	N/A	White	White	N/A
HHH	Blue Tropical	White	White	Scarlet	N/A	White	White	N/A
N	Blue Melton	Silver Bullion or Syn Met	Silver Bullion or Syn Met	Gold Syn Lace or Bullion	N/A	N/A	N/A	Gold Syn Lace or Bullion
O	Blue Melton	Silver Bullion or Syn Met	Silver Bullion or Syn Met	Scarlet	N/A	N/A	N/A	N/A
P	Blue Poplin	White	N/A	Scarlet	N/A	N/A	N/A	N/A
Q	Blue Melton	N/A	N/A	N/A	N/A	N/A	Scarlet	Scarlet
QQ	Blue Serge	N/A	N/A	N/A	N/A	N/A	Scarlet	Scarlet
QQQ	Blue Tropical	N/A	N/A	N/A	N/A	N/A	Scarlet	Scarlet
R	Bleached Twill	N/A	N/A	N/A	N/A	N/A	Green	N/A
RR	Bleached Poplin	N/A	N/A	N/A	N/A	N/A	Green	N/A

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Table II - Color and Material for Design (cont'd)

Color Combination Code Letter	Background Cloth	Rating Badge				Type of Design		
		Eagle	Specialty Mark	Chevrons	Stars	Striker Mark Specialty	Group Rate	Service Stripe
* RRR	White Polyester Twill	N/A	N/A	N/A	N/A	N/A	Green	N/A
S	Blue Melton	N/A	N/A	N/A	N/A	N/A	Green	N/A
SS	Blue Serge	N/A	N/A	N/A	N/A	N/A	Green	N/A
SSS	Blue Tropical	N/A	N/A	N/A	N/A	N/A	Green	N/A
T	Blue Melton	N/A	N/A	N/A	N/A	N/A	Blue	N/A
TT	Blue Serge	N/A	N/A	N/A	N/A	N/A	Blue	N/A
TTT	Blue Tropical	N/A	N/A	N/A	N/A	N/A	Blue	N/A
U	Bleached Twill	N/A	N/A	N/A	N/A	N/A	Blue	N/A
UU	Bleached Poplin	N/A	N/A	N/A	N/A	N/A	Blue-bird	N/A
* UUU	White Polyester Twill	N/A	N/A	N/A	N/A	N/A	Blue-bird	N/A
V	Bleached Twill	N/A	N/A	N/A	N/A	N/A	Scarlet	N/A
VV	Bleached Poplin	N/A	N/A	N/A	N/A	N/A	Scarlet	N/A
* VVV	White Polyester Twill	N/A	N/A	N/A	N/A	N/A	Scarlet	N/A

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Table II - Color and Material for Design (cont'd)

Color Combination Code Letter	Background Cloth	Rating Badge				Type of Design		
		Eagle	Specialty Mark	Chevrons	Stars	Striker Mark Specialty	Group Rate	Service Stripe
W(MC)	Blue Gabardine	White	White	Scarlet	White	White	White	N/A
X(MC)	Green Serge	Blue	Blue	Blue	Blue	Blue	Blue	Blue
* XX (MC)	Green Gabardine	Blue	Blue	Blue	Blue	Blue	Blue	Blue
Y(MC)	Green Tropical	Blue	Blue	Blue	Blue	Blue	Blue	Blue

NOTE: (a) Abbreviation N/A - Not applicable
(b) Where embroidery material is not indicated, rayon yarn is required.

3.5.2 Design.

* 3.5.2.1 Schiffli embroidered insignia. Stitch enlargements shall be furnished by the Government, to the contractor for use in preparing the necessary punchings. The stitch enlargements shall not be altered in any way. All punchings shall be approved by the Government for conformity to requirements for minimum number of stitches as specified for the applicable insignia in Table III, prior to start of production.

3.5.2.2 Hand metallic embroidered. The design of the hand metallic or synthetic metallic embroidered and/or gold lace insignia shall be in accordance with the Government approved guide sample and shall not be altered in any way.

3.5.2.3 Direction of design. Background fabrics having a twill or cord effect in their weave shall have the height dimension of the embroidered design finishing in the length (warp or wale) direction of the fabric.

3.5.3 Stitch requirements (Schiffli embroidered). The stitch requirements and dimensions used for various types of embroidered insignia shall be as specified in Table III.

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Table III - Stitch Requirements

Type and Class	Minimum No. of Stitches	Design and Dimensions 4/	Finished Sizes (inches)	Stitch Enlargement Drawing No.
* Type I, Class 1 - Chief Petty Officer	8104 1/	see Figure 1	4 X 5-7/8	16-1-861
* Type I, Class 2 - Petty Officer, First Class	6732 1/	see Figure 2	4 X 5-7/8	16-1-861
* Type I, Class 2 - Petty Officer, Second Class	5232 1/	see Figure 2	4 X 5-7/8	16-1-861
* Type I, Class 2 - Petty Officer, Third Class	3732 1/	see Figure 2	4 X 5-7/8	16-1-861
* Type I, Class 5 - Master Chief Petty Officer	8702 1/ 2/	see Figure 3	4 X 7-1/4	16-1-198 and 16-1-861
* Type I, Class 6 - Senior Chief Petty Officer	8403 1/ 2/	see Figure 3	4 X 7-1/4	16-1-198 and 16-1-861
* Type I, Class 7 - Master Chief Petty Officer of the Navy	9001 1/ 2/	see Figure 3	4 x 7-1/4	16-1-198 and 16-1-861
Type II, Specialty Mark or Striker Mark 3/				
Aerographer's Mate	995	see Figure 4	2-7/8X1-7/8	16-1-283
Airman	1305	see Figure 4	3-3/16x 1-11/16	16-1-579
Air Controlman	1050	see Figure 4	2-7/8X1-7/8	16-1-207
Aircrew Survival Equipmentman	1075	see Figure 4	2-7/8x1-7/8	16-1-23
Aviation Structural Mechanic	910	see Figure 4	2-7/8x1-7/8	16-1-291
Aviation Antisubmarine Warfare Technician	965	see Figure 4	2-7/8x1-7/8	16-1-529
Aviation Boatswain's Mate	935	see Figure 4	2-7/8X1-7/8	16-1-285

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Table III - Stitch Requirements (cont'd)

Type and Class	Minimum No. of Stitches	Design and Dimensions 4/	Finished Sizes (inches)	Stitch Enlargement Drawing No.
Aviation Electrician's Mate	895	see Figure 4	2-7/8X1-7/8	16-1-286
Aviation Electronics Technician	825	see Figure 5	2-7/8X1-7/8	16-1-208
Aviation Fire Control Technician	1250	see Figure 5	2-7/8X1-7/8	16-1-287
Aviation Machinist's Mate	960	see Figure 5	2-7/8x1-7/8	16-1-288
Aviation Maintenance Administrationman	1083	see Figure 5	2-7/8x1-7/8	16-1-289
Aviation Ordnanceman	1095	see Figure 5	2-7/8x1-7/8	16-1-206
Aviation Storekeeper	970	see Figure 5	2-7/8X1-7/8	16-1-290
Aviation Antisubmarine Warfare Operator	1346	see Figure 5	2-1/2x1-3/4	16-1-531
Aviation Support Equipment Technician	867	see Figure 5	2-7/8x1-7/8	16-1-292
Boatswain's Mate	580	see Figure 6	2x2	16-1-293
Boiler Technician	745	see Figure 6	2x2	16-1-295
Builder	675	see Figure 6	2x2	16-1-296
Cryptologic Technician	405	see Figure 6	2x2	16-1-209
Construction Electrician	465	see Figure 6	2x2	16-1-298
Construction Mechanic	610	see Figure 6	2x2	16-1-299
Damage Controlman	405	see Figure 6	2x2	16-1-442
Data Processing Technician	550	see Figure 6	2x2	16-1-24
Data Systems Technician	597	see Figure 7	2x2	16-1-300
Dental Technician	755	see Figure 7	2x2	16-1-301
Disbursing Clerk	574	see Figure 7	2x2	16-1-532

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Table III - Stitch Requirements (cont'd)

Type and Class	Minimum No. of Stitches	Design and Dimensions 4/	Finished Sizes (inches)	Stitch Enlargement Drawing No.
Electrician's Mate	1280	see Figure 7	2x2	16-1-204
Electronics Technician	450	see Figure 7	2x2	16-1-221
Electronic Warfare Technician	489	see Figure 7	2x2	16-1-254
Engineering Aid	447	see Figure 7	2x2	16-1-530
Engineman	690	see Figure 7	2x2	16-1-303
Equipment Operator	790	see Figure 7	2x2	16-1-305
* Fire Controlman	740	see Figure 8	1-7/16x5/8	16-1-625
Fireman	1354	see Figure 8	2-5/16x2-1/8	16-1-580
Fire Control Technician	540	see Figure 8	2x2	16-1-306
Gas Turbine Systems Technician	543	see Figure 8	2x2	16-1-466
Gunner's Mate	515	see Figure 8	2x2	16-1-307
Hospital Corpsman	685	see Figure 8	2x2	16-1-203
Hull Maintenance Technician	463	see Figure 8	2x2	16-1-252
I.C. Electrician	1160	see Figure 8	2x2	16-1-308
Illustrator Draftsman	590	see Figure 8	2x2	16-1-309
Intelligence Specialist	358	see Figure 8	2x2	16-1-498
Instrumentman	305	see Figure 9	2x2	16-1-310
Journalist	460	see Figure 9	2x2	16-1-311
Legalman	485	see Figure 9	2x2	16-1-389
Lithographer	450	see Figure 9	2x2	16-1-312
Machinist's Mate	740	see Figure 9	2x2	16-1-205
Machinery Repairman	490	see Figure 9	2x2	16-1-313

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Table III - Stitch Requirements (cont'd)

Type and Class	Minimum No. of Stitches	Design and Dimensions 4/	Finished Sizes (inches)	Stitch Enlargement Drawing No.
Master-At-Arms	703	see Figure 9	1-1/2X1-1/2	16-1-460
Master C.P.O. of the Fleet Command	344	see Figure 9	1-1/2x1-1/2	16-1-450
Mess Management Specialist	705	see Figure 9	2x2	16-1-527
Mineman	680	see Figure 10	2x2	16-1-314
Missile Technician	425	see Figure 10	2x2	16-1-315
Molder	645	see Figure 10	2x2	16-1-316
Musician	510	see Figure 10	2x2	16-1-317
Navy Counselor	548	see Figure 10	1-1/2X1-1/2	16-1-461
Ocean Systems Technician	643	see Figure 10	1-7/8X1-1/2	16-1-202
Operations Specialist	410	see Figure 10	2x2	16-1-324
Opticalman	900	see Figure 10	2x2	16-1-318
Patternmaker	605	See Figure 10	2x2	16-1-319
Personnelman	745	see Figure 11	2x2	16-1-320
Photographer's Mate	405	see Figure 11	2x2	16-1-321
Postal Clerk	751	see Figure 11	2x2	16-1-322
Quartermaster	590	see Figure 11	2x2	16-1-323
Radioman	625	see Figure 11	2x2	16-1-325
Religious Program Specialist	1029	see Figure 11	2-1/4X1-5/8	16-1-578
Signalman	575	see Figure 11	2x2	16-1-328
Ship's Serviceman	355	see Figure 11	2x2	16-1-327
Seaman	936	see Figure 12	2-3/8x2-3/8	16-1-581
Sonar Technician	525	see Figure 12	2-7/8X1-7/8	16-1-528

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Table III - Stitch Requirements (cont'd)

Type and Class	Minimum No. of Stitches	Design and Dimensions 4/	Finished Sizes (inches)	Stitch Enlargement Drawing No.
Steel Worker	605	see Figure 12	2x2	16-1-330
Storekeeper	550	see Figure 12	2x2	16-1-332
Torpedoman's Mate	825	see Figure 12	2-7/8x1-7/8	16-1-333
Tradevman	812	see Figure 12	2x2	16-1-526
Utilitiesman	565	see Figure 12	2x2	16-1-335
* Weapons Technician	606	see Figure 12	1x1	16-1-852
Yeoman	300	see Figure 12	2x2	16-1-210
Type V - Group Rate Mark				
Two Stripes (class 2)	1780	see Figure 13	3-1/16x 2-15/16	16-1-15
Three Stripes (class 3)	2670	see Figure 13	3-1/16x3-5/8	16-1-15
Type VI - Stripe, Service				
Class A				
One Stripe	2384	see Figure 14	7-3/4x7/8	16-1-378
Two Stripes	4768	see Figure 14	7-3/4x1-1/2	16-1-378
Three Stripes	7152	see Figure 14	7-3/4x2-1/8	16-1-378
Four Stripes	9536	see Figure 14	7-3/4x2-3/4	16-1-378
Five Stripes	11920	see Figure 14	7-3/4x3-3/8	16-1-378
Six Stripes	14304	see Figure 14	7-3/4x4	16-1-378
Seven Stripes	16688	see Figure 14	7-3/4x4-5/8	16-1-378
Class B (Rayon)				
One Stripe	2247	see Figure 14	6x7/8	16-1-447
Two Stripes	4494	see Figure 14	6x1-1/2	16-1-447
Three Stripes	6741	see Figure 14	6x2-1/8	16-1-447
Four Stripes	8988	see Figure 14	6x2-3/4	16-1-447

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Table III - Stitch Requirements (cont'd)

Type and Class	Minimum No. of Stitches	Design and Dimensions 4/	Finished Sizes (inches)	Stitch Enlargement Drawing No.
Five Stripes	11235	see Figure 14	6x3-3/8	16-1-447
Six Stripes	13482	see Figure 14	6x4	16-1-447
Seven Stripes	15729	see Figure 14	6x4-5/8	16-1-447
* Class C				
One Stripe	2046	see Figure 14	7-3/4x7/8	16-1-865
Two Stripes	4092	see Figure 14	7-3/4x1-1/2	16-1-865
Three Stripes	6138	see Figure 14	7-3/4x2-1/8	16-1-865
Four Stripes	8184	see Figure 14	7-3/4x2-3/4	16-1-865
Five Stripes	10230	see Figure 14	7-3/4x3-3/8	16-1-865
Six Stripes	12276	see Figure 14	7-3/4x4	16-1-865
Seven Stripes	14322	see Figure 14	7-3/4x4-5/8	16-1-865
* Class D				
One Stripe	1725	see Figure 14	6x7/8	16-1-863
Two Stripes	3450	see Figure 14	6x1-1/2	16-1-863
Three Stripes	5175	see Figure 14	6x2-1/8	16-1-863
Four Stripes	6900	see Figure 14	6x2-3/4	16-1-863
Five Stripes	8625	see Figure 14	6x3-3/8	16-1-863
Six Stripes	10350	see Figure 14	6x4	16-1-863
Seven Stripes	12075	see Figure 14	6x4-5/8	16-1-863
Type VIII - Insignia, Grade	299	see Figure 15	-	16-1-198
Type X - Rating Badge Utility, P01	6980	see Figure 16	3-3/4x5-1/4	16-1-592

1/ Exclusive of the number of stitches for the Specialty mark.
2/ Minimum number of stitches for one, two, or three stars as applicable.

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3/ Finished sizes are applicable to Striker Mark only.
4/ Figures 1 thru 16 are for dimensions only and are not drawn to scale. For correct design and proper stitch execution refer to the applicable stitch enlargement. Unless otherwise specified, all dimensions are in inches.

3.6 Type I, Badge Rating, Classes 1, 2, 5, 6, and 7.

* 3.6.1 Construction (Schiffli embroidered). Type I, Classes 1, 2, 5, 6, and 7 rating badges shall be embroidered on the background cloth specified in Table II in accordance with the applicable stitch enlargement listed in Table III, the dimensions specified in the applicable figure, and the construction requirements of 3.4.1.

3.6.2 Construction (hand embroidered). Type I, classes 1, 2, 5, 6, and 7 rating badges shall be embroidered on the background cloth specified in Table II in accordance with an approved guide sample, the dimensions specified in the applicable figure, and the construction requirements of 3.4.2.

3.6.3 Type II - Specialty Mark and Striker Mark. The specialty mark shall be centered on the rating badge between the upper chevron and the eagle's talons. Striker mark shall be centered on the background cloth specified in Table II and shall be embroidered in accordance with the designs and dimensions indicated for specialty marks and striker marks in Figures 4 thru 12. Construction shall be in accordance with the applicable requirements of 3.6.1 or 3.6.2.

* 3.6.4 Type V, Group Rate Mark. The group rate marks shall consist of two (class 2) stripes or three (class 3) stripes embroidered diagonally on the background cloth specified in Table II. The stripes shall be Schiffli embroidered in accordance with the applicable stitch enlargement list in Table III, the dimensions specified in Figure 13, and the construction requirements of 3.4.1.

* 3.6.5 Type VI, Stripe, Service, Class A, B, C and D. Class A, B, C and D insignia shall consist of one thru seven stripes, as applicable, embroidered diagonally on the background cloth specified in Table II. The stripes shall be Schiffli embroidered in accordance with the applicable stitch enlargement listed in Table III, the dimensions specified in Figure 14, and the construction requirements of 3.4.1.

3.6.5.1 Gold lace service stripes. Insignia shall be produced in accordance with 3.4.3 and the dimensions specified in Figure 14.

* 3.6.6 Type VIII - Insignia grade (Star). The star(s) shall be of a 5-point design in accordance with the dimensions specified in Figure 15. Stars(s) shall be Schiffli embroidered in accordance with 3.4.1 and 3.6.1 or hand embroidered in accordance with 3.4.2 and 3.6.2.

3.6.7 Type X, Badge, Rating, Utility, Petty Officer, First Class. The unisized rating badge shall be embroidered on the background cloth specified in Table II in accordance with the applicable stitch enlargement listed in Table III, the dimensions specified in the applicable figure, and the construction requirements of 3.4.1.

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3.7 Measurement and Positioning of Tolerances. The tolerances listed in Table IV shall be applicable to specified dimensions for finished insignia.

Table IV - Tolerances

Inches		Tolerance(inch)
Dimensions	under 15/16	± 1/32
Dimensions ranging -	from 1 to 1-15/16	± 1/16
	from 2 to 2-15/16	± 3/32
	from 3 to 3-15/16	± 1/8
	from 4 to 4-15/16	± 5/32
	from 5 to 5-15/16	± 3/16
	from 6 to 6-15/16	± 7/32
	from 7 to 7-15/16	± 1/4
Degrees		± 3

3.8 Workmanship. The finished insignia shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. The first article, submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship and dimensional requirements.

* 4.3.1 Approval of punchings. Before production is commenced, punchings for the applicable insignia shall be inspected and approved by the Government for conformance to the requirements for minimum number of stitches specified in Table III.

* 4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. A certificate of compliance will be accepted for the labile sulfur requirements of 3.3.16. In addition, components and materials listed in Table V shall be tested for the characteristics shown following methods specified in FED-STD-191, wherever applicable. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the lot average or one or more sample units fail to meet the requirement specified. The sample unit and sample size shall be as follows:

Component	Unit of Product Inspection Lot	Unit of Product Sample
Buckram	yards	1 linear yard
Embroidery yarn	holders	1 cone or spool, or sufficient number of bobbins to total a minimum of 360 yards
Thread (all types and sizes)	holders	1 cone or spool, or sufficient number of bobbins to total a minimum of 360 yards (when threads are purchased on cones, spools and on bobbins, each shall be tested.)

The sample size for each component shall be as follows:

Lot Size (Units)	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table V - Component Testing

Component	Characteristics	Requirement Paragraph	Test Method	No. of Determinations Per Unit of Product	Results Reported As
Buckram (Types A and B)	Mat'l. Ident.1/	3.3.13	1/	1	Pass or Fail
	Yarn count				
	Ends	3.3.13	5050	5	Pass or Fail
	Picks	3.3.13	5050	5	Pass or Fail
	Weight	3.3.13	5041	5	Pass or Fail
	Weave	3.3.13	Visual	1	Pass or Fail
Rayon Embroidery yarn	Material ident	3.3.14.1	1/	1	Pass or Fail
	Denier	3.3.14.1	4021	1	Pass or Fail
	Ply	3.3.14.1	Visual	1	Pass or Fail
	Presence of sulfur	3.3.14.1.1	2020 1/	1	Pass or Fail
Bobbin Thread	Material ident	3.3.14.2	1/	1	Pass or Fail
	Ply	3.3.14.2	Visual	1	Pass or Fail
Thread, cotton	Colorfastness to Dry heat (hot pressing)	3.3.15.1	5640	1	Pass or Fail
	Wet heat (hot pressing)	3.3.15.1	5640	1	Pass or Fail

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

* 4.4.2 Examination of the end item. The defects found during the examination of the end item shall be classified in accordance with 4.4.2.1 and 4.4.2.2. The sample unit shall be one finished insignia. When insignia are furnished in pairs, selection shall be by pairs. The inspection levels and acceptable quality levels shall be as specified in 4.4.2.3. The lot size shall be expressed in terms of the sample unit.

4.4.2.1 Visual examination. The finished insignia shall be examined for defects in design, material, construction and workmanship and classified in accordance with the defects listed in Table VI.

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Table VI - Classification of Defects

Examine	Defects
I Material	a. Other than specified.
	b. Color or shade other than specified.
	c. Hole, float, slub, knot, broken pick or end, spot or stain (i.e., background cloth).
II Design	a. Any deviation from applicable stitch enlargement drawing, illustration, or guide sample.
	b. Wrong color or wrong color arrangement.
	c. Not centered in the prescribed background.
	d. Form board used to raise design missing (when applicable).
III Construction	a. Stitching incomplete.
	b. Embroidered on wrong side of background cloth.
	c. Broken or skipped stitches.
	d. Shift, affecting appearance.
	e. Incomplete coverage, background cloth or underlay stitches visible.
	f. Density not in accordance with guide sample.
	g. Bobbin threads visible, affecting appearance.
	h. Bunched or tangled threads and/or yarns.
	i. Loose or tight tension affecting appearance.
	j. Puckering exceeding guide sample (i.e., Specialty Mark or Rating Badge).
	k. Stitches out of alignment affecting appearance.
	l. Open or poorly formed points, (i.e., gold lace rating badge).
	m. Unevenly spaced, (i.e., gold lace rating badge and gold lace service stripes).

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Table VI - Classification of Defects (cont'd)

Examine	Defects
	n. Buckram omitted, incomplete, or extending beyond insignia, (i.e., metallic embroidered, synthetic metallic embroidered and gold lace insignia).
	o. Height dimension of embroidered design not finishing in length direction of (when applicable) background cloth having a twill or cord effect in its weave.
IV Finish	a. Stitches per inch not within specified range.
	b. Tight or loose tension affecting appearance.
	c. Irregular or unevenly gauged stitching, affecting appearance.
	d. Broken or skipped stitches extending more that 1/4 inch.
	e. Color of hemming thread not as specified.
	f. Ends not fitted into slits cut in background cloth, (i.e., gold lace chevrons and service stripes).
	g. Points not mitered, (i.e., gold lace chevrons on rating badge).
	h. Edges not finished as required.
	i. Wrong stitch type.
	j. Wrong seam type.
	k. Stitching run-off.
V Workmanship	a. Spot or stain.
	b. Misshaped or distorted.
	c. Two or more thread ends longer than 1/16 inch.
	d. Color of insignia face affected by stiffener coating.

4.4.2.2 Dimensional examination. The finished insignia shall be examined for defects in dimension. Any measurement deviating from dimensions or tolerances as specified on applicable figure and in Table IV shall be scored a defect.

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4.4.2.3 Inspection levels and acceptable quality levels. The inspection levels and acceptable quality levels (AQL's), expressed in defects per hundred units, for visual and dimensional examinations shall be as follows:

	Inspection Level	AQL
For defects applicable to 4.4.2.1	II	4.0
For defects applicable to 4.4.2.2	S-3	6.5

* 4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing and marking comply with the Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, not as specified or damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling; bulging or distortion of containers.
Content	Number of items per container is more or less than specified; wrong type, class or style per set-up box; number of set-up boxes per shipping container more or less than specified.

4.4.4 Testing of the end item. The end item shall be tested as specified in Table VII. The lot size shall be expressed in units of one insignia of each type or class. The unit of product for tests shall be six insignia. The lot shall be unacceptable if one or more sample units fail to meet the requirements referenced in Table VII. The sample size shall be as follows:

Lot Size (each)	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table VII - End Item Tests

Characteristic	Requirement Paragraph	Test Method	No. Per Unit of Product	Determinations Results Reported as
Rayon yarn Colorfastness to: 1/ 2/		3/		
Bleaching	3.3.14.1.1.1	5600	1	Pass or Fail
Dry Heat(hot pressing)	3.3.14.1.1.1	5640	1	Pass or Fail
Wet Heat(hot pressing)	3.3.14.1.1.1	5640	1	Pass or Fail
Light	3.3.14.1.1.1	5660	1	Pass or Fail
Laundering	3.3.14.1.1.1	5614	1	Pass or Fail
Wet-dry cleaning	3.3.14.1.1.1	5622	1	Pass or Fail
Shade Match of Embroidered Yarn	3.3.14.1.2	Visual	1 of each color	Pass or Fail
Type of Base Cloth	3.3.1 Thru 3.3.10	Visual	1	Pass or Fail

1/ Test reports will be furnished for each color in the initial shipping lot of the finished insignia. A certificate of compliance will be accepted for each color for each subsequent lot on the same contract, provided the contractor certifies that the dyer or dye formulation has not been changed.

2/ When the base cloth specification does not specify a colorfastness test for laundering Table VII requirements for fastness to bleaching and laundering shall not apply.

3/ Except that the test specimens shall be the finished embroidered insignia.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or C (see 6.2).

5.1.1 Level A.

5.1.1.1 Quantity per set-up box.

5.1.1.1.1 Type I - Badge, Rating - All Classes. Each rating badge shall be individually packaged in a properly fitted flat style polyethylene bag conforming to the requirements specified in 5.1.1.2. The bag opening shall be across the face-width. The open end of the bag shall be folded over. Twenty (20) rating badges of one class and rate only shall be packaged in a set-up box that conforms to the requirements specified in 5.1.1.3.1.

5.1.1.1.2 Type II - Specialty Mark and Striker Mark. Ten (10) marks, of one kind only, shall be packaged in a properly fitted flat style polyethylene bag conforming to the requirements specified in 5.1.1.2. The bag opening shall be across the face-width. The open-end of the bag shall be folded over. One hundred and twenty (120) marks (12 bags of 10 each), of one kind only shall be packaged in a set-up box that conforms to the requirements specified in 5.1.1.3.1.

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5.1.1.1.3 Type V - Rate Mark - All Classes. Ten (10) marks, of one kind only, shall be packaged in a properly fitted flat style polyethylene bag conforming to the requirements specified in paragraph 5.1.1.2. The bag opening shall be across the face width. The open end of the bag shall be folded over. Sixty (60) marks (6 bags of 10 each) shall be packaged in a set-up box that conforms to the requirements specified in 5.1.1.3.1.

5.1.1.1.4 Type VI -Stripe, Service - All Classes.

240 each (1 stripe insignia)
150 each (2 stripe insignia)
90 each (3 stripe insignia)
60 each (4 or 5 stripe insignia)
30 each (6 or 7 stripe insignia)

The above quantities of service stripes, of one kind only, shall be packaged in a set-up box that conforms to the requirements specified in 5.1.1.3.2. Thirty (30) items of one kind only, shall be tied at each end by elastic band, twine or cotton tape to form a bundle.

5.1.1.1.5 Type VIII - Insignia, Grade. The insignia shall be packaged in accordance with standard industry practice.

5.1.1.1.6 Type X, Badge, Rating, Utility, Petty Officer, First Class. Each rating badge shall be individually packaged in a properly fitted flat style polyethylene bag conforming to the requirements specifed in paragraph 5.1.1.2. The bag opening shall be across the face-width and the open end shall be folded over. Twenty (20) rating badges shall be packaged in a set-up box that conforms to the requirements specified in paragraph 5.1.1.3.1.

5.1.1.2 Bags, polyethylene. The polyethylene bags shall be fabricated from clear polyethylene film of 0.0010 inch thick (± 20% tolerance). The polyethylene bags shall be formed with heat-sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag.

5.1.1.3 Boxes set-up.

5.1.1.3.1 Types I, II, V, and X Insignia. Insignia, of one type and kind only, shall be placed in a two-piece, full telescope style, paperboard set-up box conforming to type I, variety 1, class A, style 4 of PPP-B-676. The outside dimensions of the set-up box shall be 7 inches in length, 5 inches in width, and 2-1/8 inches in depth.

5.1.1.3.2 Type VI Insignia. Insignia, of one class and kind only shall be placed in a two-piece, full telescope style, set-up paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. The outside dimensions of the set-up box shall be 11-1/2 inches in length, 7-5/8 inches in width and 2-3/8 inches in depth. Unless otherwise specified, the interior of set-up boxes for items containing wool shall be sprinkled with not less than 1/4 ounce of flaked naphthalene.

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NOTE: The naphthalene flake requirement is not applicable when Government furnished material with wool or part wool content has been mothproof treated.

5.1.2 Level C. Insignia shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B or C (see 6.2).

5.2.1 Level A.

5.2.1.1 Types I, II, V, and X Insignia. Fifteen (15) set-up boxes containing one type and class of insignia packaged as specified in 5.1.1, shall be packed in a shipping container conforming to type CF, size 1, class weather-resistant, variety DW, grade V15c of MIL-B-17757. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of the box specification.

5.2.1.2 Type VI Insignia. Six (6) set-up boxes containing one class of insignia packaged as specified in 5.1.1, shall be packed in a shipping container conforming to type CF, size 1, class weather-resistant, variety DW, grade V15c. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of the box specification.

5.2.2 Level B.

5.2.2.1 Types I, II, V, and X Insignia. Fifteen (15) set-up boxes containing one type and class of insignia, packaged as specified in 5.1.1, shall be packed in a shipping container conforming to type CF, size 1, class domestic, variety DW, grade 200 of MIL-B-17757. Shipping containers shall be closed in accordance with the appendix of the box specification.

5.2.2.2 Type VI Insignia. Six (6) set-up boxes containing one class of insignia, packaged as specified in 5.1.1, shall be packed in a shipping container conforming to type CF, size 1, class domestic, variety DW, grade 200 of MIL-B-17757. Shipping containers shall be closed in accordance with the appendix of the box specification.

5.2.3 Level C. Finished insignia, packaged as specified in 5.1, shall be packed in a manner to insure safe delivery at destination at the lowest transportation rate for such deliveries. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

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* 5.4 MIL-STD-2073. When specified (see 6.2) Preservation and Packaging shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The embroidered insignia are intended to designate rate and specialty for enlisted men of the Navy and Naval personnel attached to Marine Corps units.

* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
c. Type, class, and color combination required (see 1.2 and 3.5.1).
d. When first article inspection is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quality, testing and approval of the first article.
e. Cable number for yarn (see 3.3.14.1.1).
f. Levels of preservation and packing required (see 5.1 and 5.2)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.4).

6.3 Samples. For access to sample of the end item and shade sample, address the procuring activity issuing the invitation for bids.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only the the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Subject term (key word) listing.
Badge, rating
Mark, rate, group
Mark, specialty and striker
Personnel, enlisted

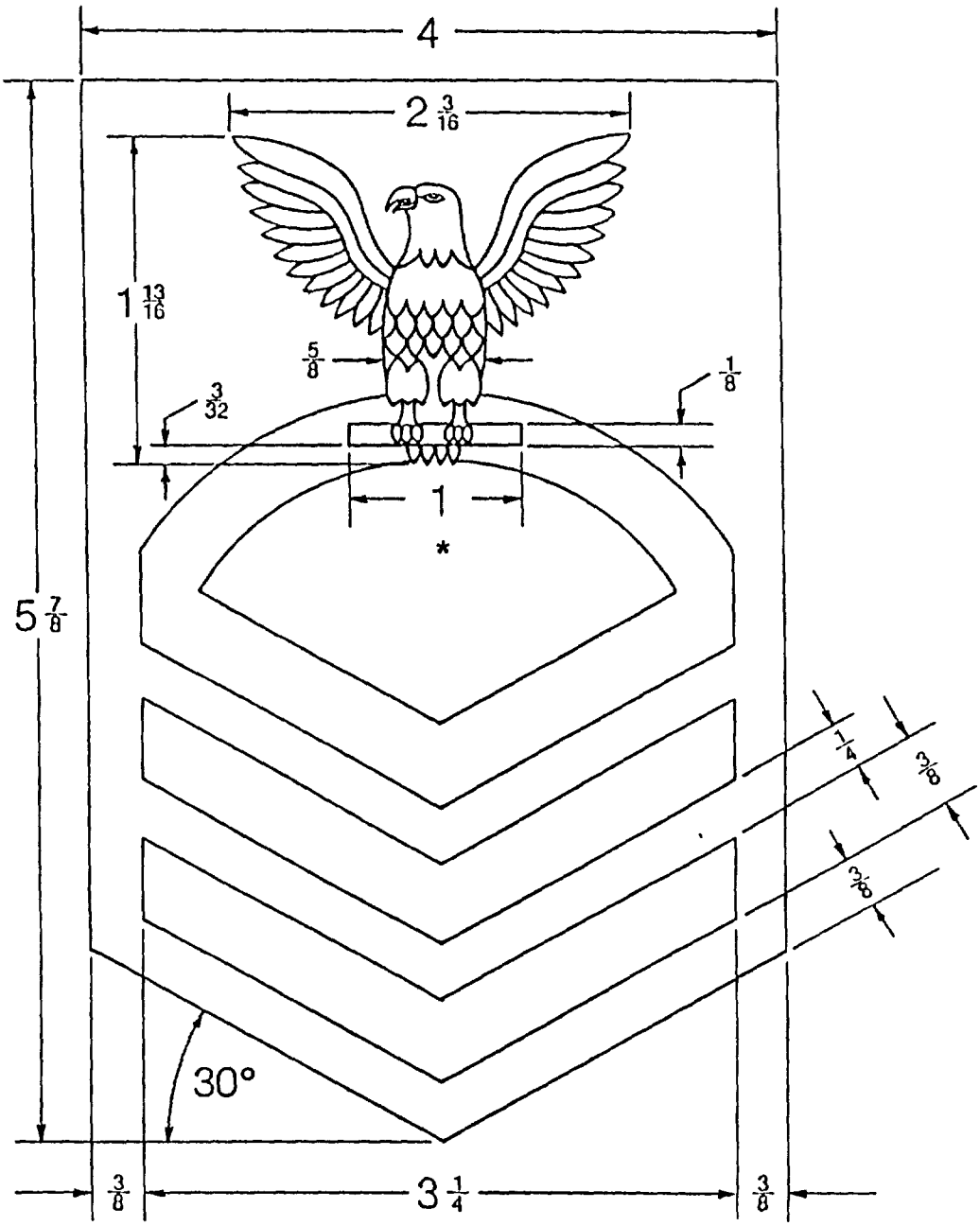
Custodian:
Navy - NU

Review activities:
Navy - MC
DLA - CT

Preparing activity:
Navy - NU

Project 8455-N446

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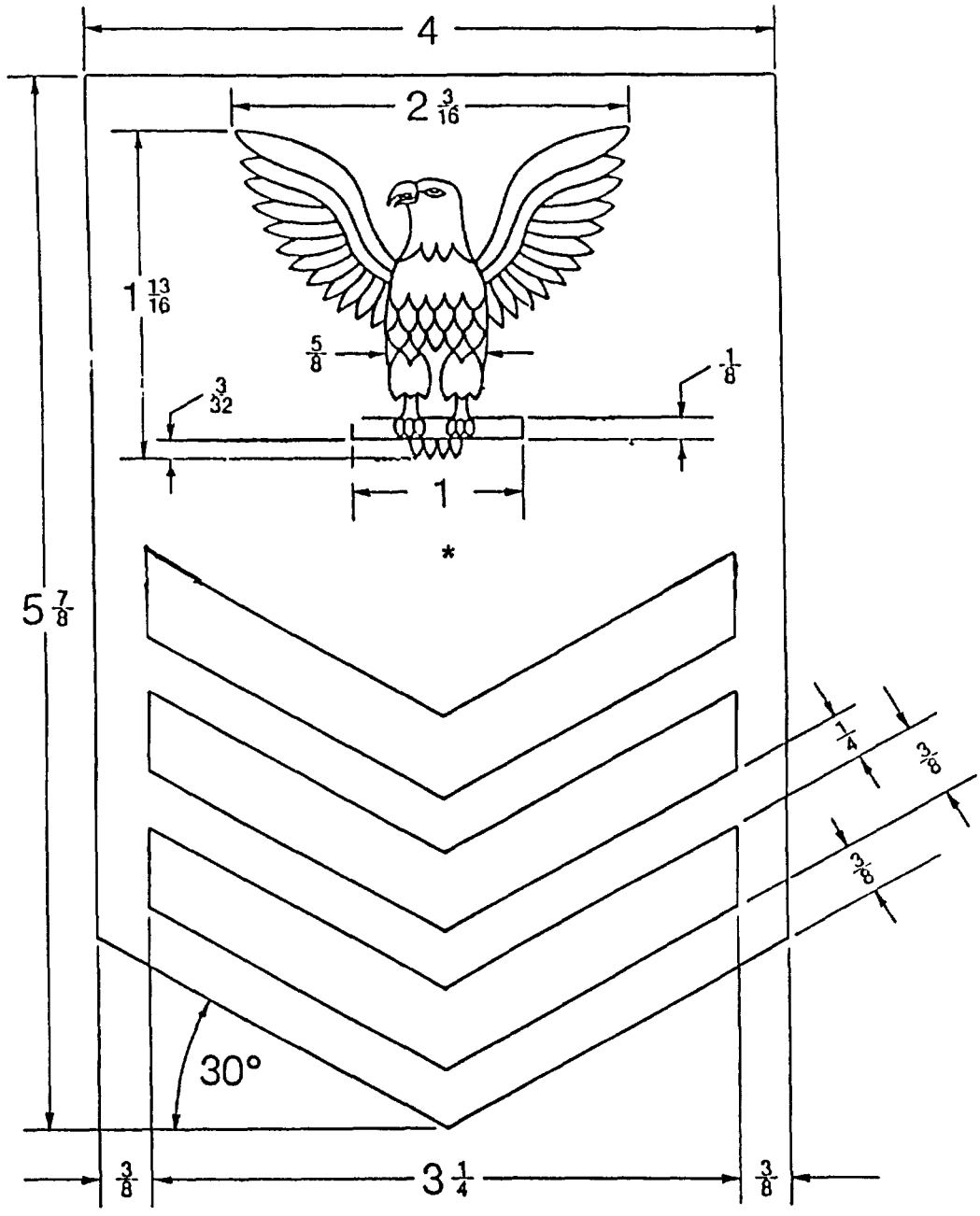


*Applicable Specialty Mark - Optically Centered

NOTE: ILLUSTRATION IS FOR DIMENSIONS ONLY. FOR CORRECT DESIGN AND PROPER STITCH EXECUTION REFER TO THE APPLICABLE STITCH ENLARGEMENT. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

Figure 1 - Type I, Class 1, Chief Petty Officer Rating Badge

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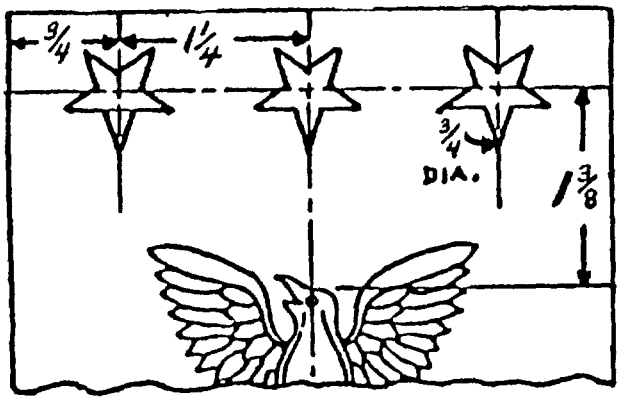


*Applicable Specialty Mark -
Optically Centered

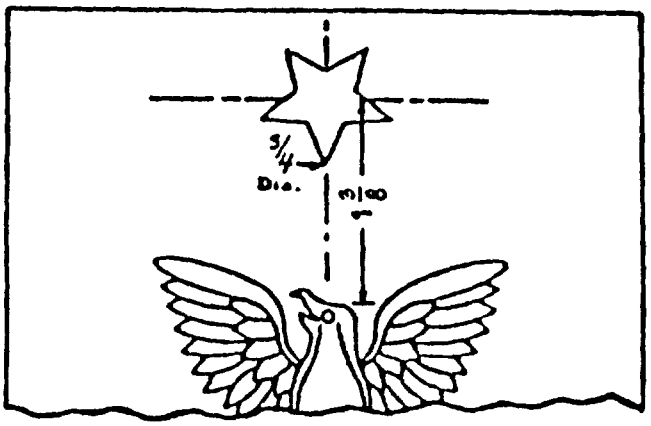
NOTE: ILLUSTRATION IS FOR DIMENSIONS ONLY. FOR CORRECT DESIGN AND PROPER STITCH EXECUTION REFER TO THE APPLICABLE STITCH ENLARGEMENT. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

Figure 2 - Type I, Class 2, Chief Petty Officer First Class Rating Badge

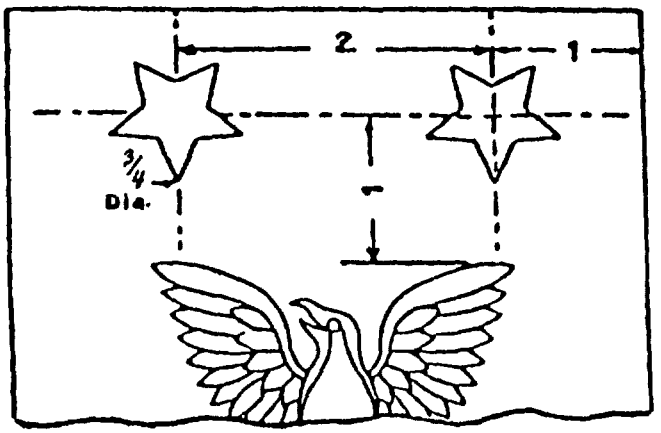
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:06Z
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TYPE I, CLASS 7 - MASTER CHIEF PETTY OFFICER OF THE NAVY



TYPE I, CLASS 6 - SENIOR. CHIEF PETTY OFFICER

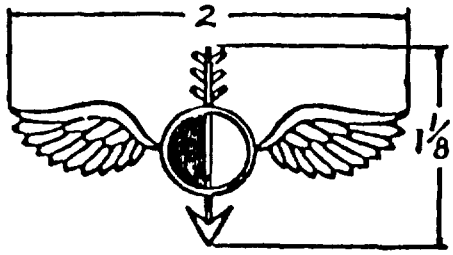


TYPE I, CLASS 5 - MASTER. CHIEF PETTY OFFICER

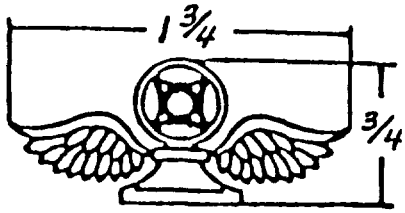
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Figure 3. Type I, Class 7, Master Chief Petty Officer; Type I, Class 6, Senior Chief Petty Officer; Type I, Class 5, Master Chief Petty Officer

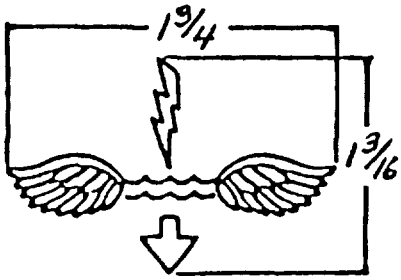
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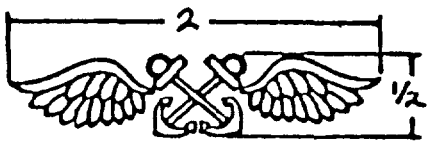
AEROGRAPHER'S MATE
(AG)



AIR CONTROLMAN
(AC)

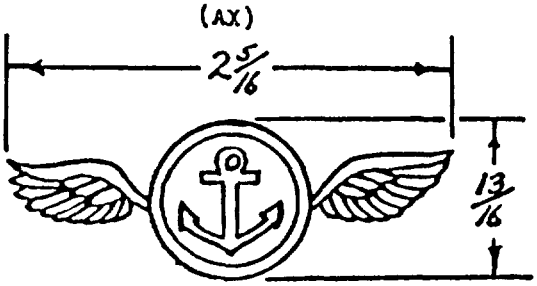


AVIATION ANTISUBMARINE WARFARE TECHNICIAN
(AX)

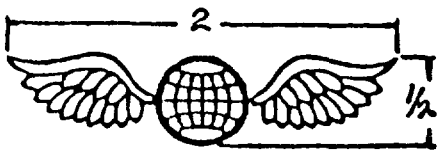


AVIATION BOATSWAIN'S MATE

(AB)

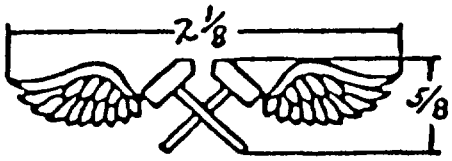


AIRMAN *



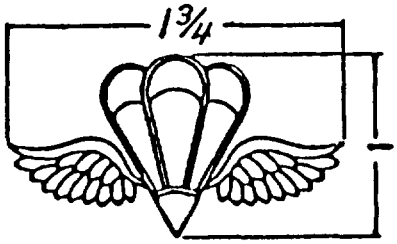
AVIATION ELECTRICIAN'S MATE

(AE)



AVIATION STRUCTURAL MECHANIC

(AM)



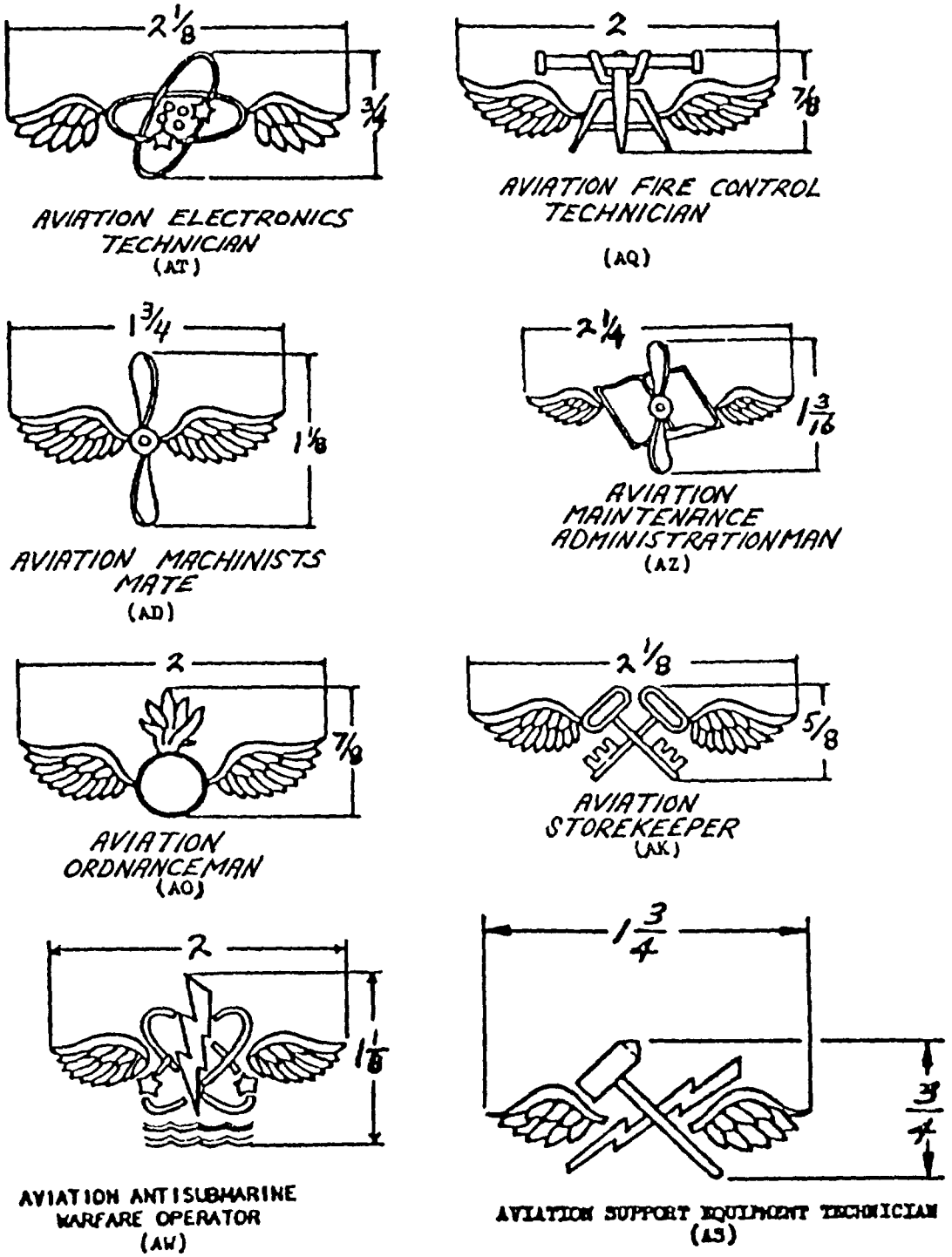
AIRCREW SURVIVAL EQUIPMENTMAN (PR)

* Not applicable to Rating Badge

NOTE: ILLUSTRATION IS FOR DIMENSIONS ONLY. FOR CORRECT DESIGN AND PROPER STITCH EXECUTION REFER TO THE APPLICABLE STITCH ENLARGEMENT. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

Figure 4 - Type II - Specialty Mark and Striker Mark

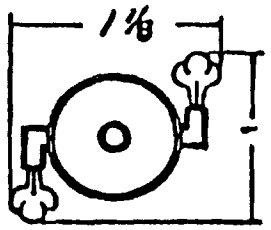
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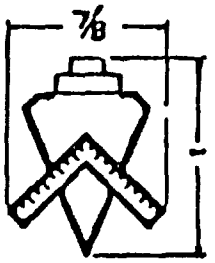
NOTE: ILLUSTRATION IS FOR DIMENSIONS ONLY. FOR CORRECT DESIGN AND PROPER STITCH EXECUTION REFER TO THE APPLICABLE STITCH ENLARGEMENT. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

Figure 5 - Type II - Specialty Mark and Striker Mark

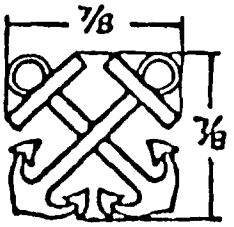
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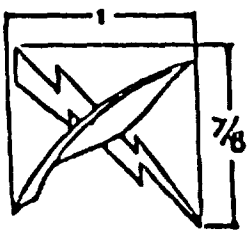
BOILER TECHNICIAN (BT)



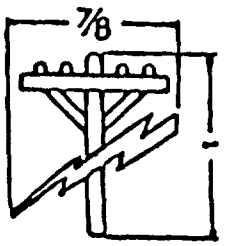
BUILDER
(BU)



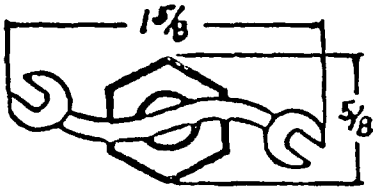
BOATSWAIN'S MATE
(BM)



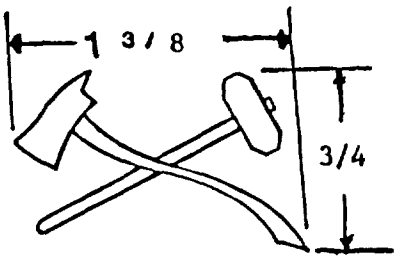
CRYPTOLOGIC TECHNICIAN
(CT)



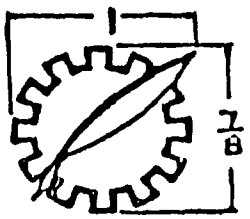
CONSTRUCTION ELECTRICIAN
(CE)



CONSTRUCTION MECHANIC
(CM)



DAMAGE CONTROLMAN(DC)

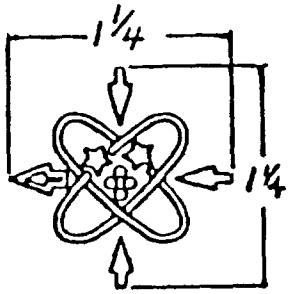


DATA PROCESSING TECH (DP)

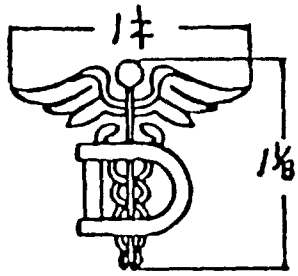
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Figure 6 - Type II - Specialty Mark and Striker Mark

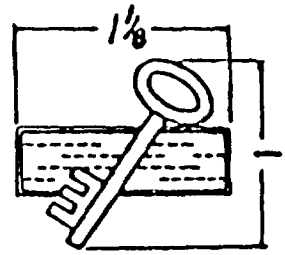
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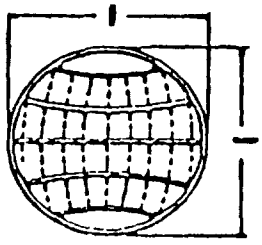
DATA SYSTEMS TECHNICIAN (DS)



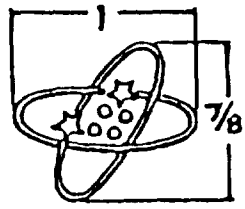
DENTAL TECHNICIAN (DT)



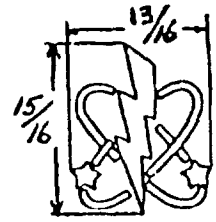
DISBURSING CLERK (DK)



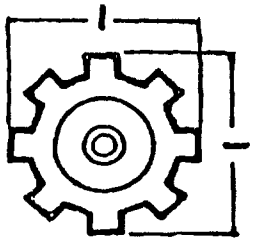
ELECTRICIAN'S MATE (EM)



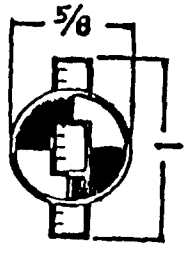
ELECTRONIC'S TECHNICIAN (ET)



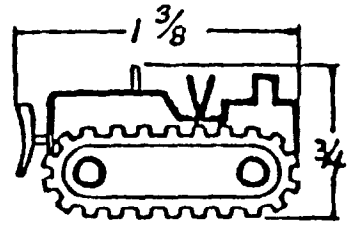
ELECTRONIC WARFARE TECH (EW)



ENGINEMAN (EN)



ENGINEERING AID (EA)

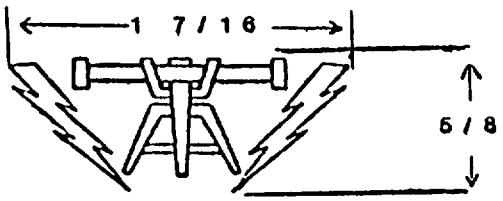


EQUIPMENT OPERATOR (EO)

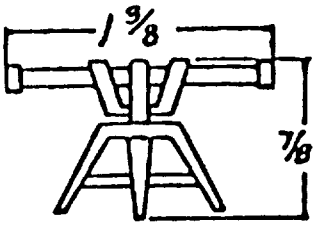
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Figure 7 - Type II - Specialty Mark and Striker Mark

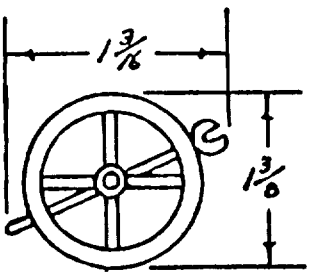
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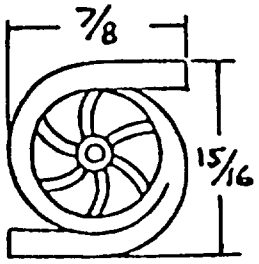
FIRE CONTROLMAN
(FC)



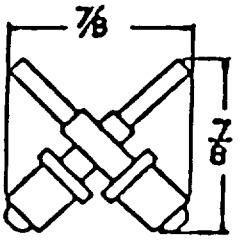
FIRE CONTROL
TECHNICIAN
(FT)



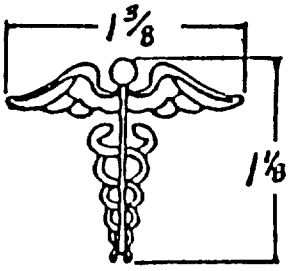
FIREMAN *



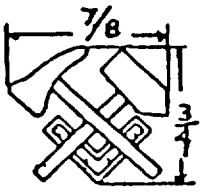
Gas Turbine
Systems Technician (GS)



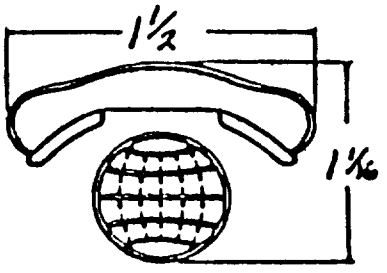
GUNNER'S MATE
(GM)



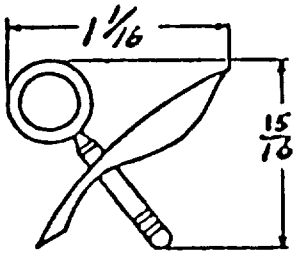
HOSPITAL
CORPSMAN
(HM)



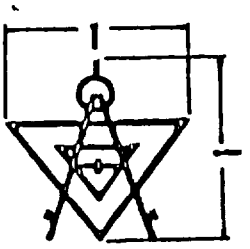
HULL MAINTENANCE TECH (HT)



I.C. Electrician (I.C.)



INTELLIGENCE SPECIALIST (IS)



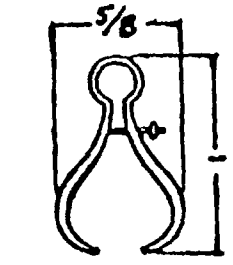
ILLUSTRATOR
DRAFTSMAN
(DM)

* Not applicable to Rating Badge

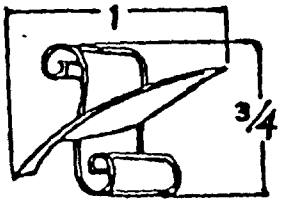
NOTE: ILLUSTRATION IS FOR DIMENSIONS ONLY. FOR CORRECT DESIGN AND PROPER STITCH EXECUTION REFER TO THE APPLICABLE STITCH ENLARGEMENT. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

Figure 8 - Type II - Specialty Mark and Striker Mark

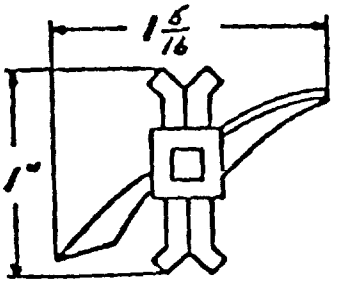
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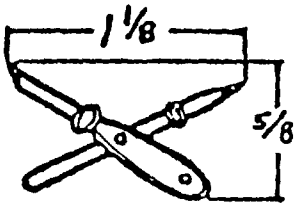
INSTRUMENTMAN
(IM)



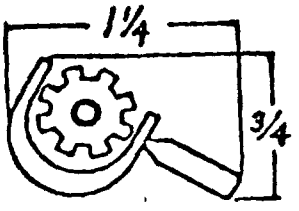
JOURNALIST
(JO)



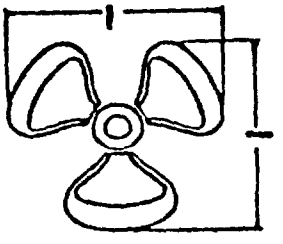
LEGALMAN (LN)



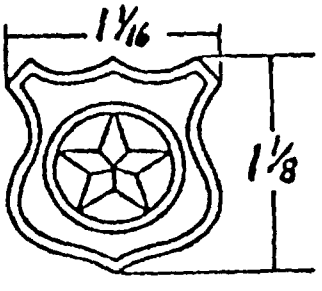
LITHOGRAPHER
(LI)



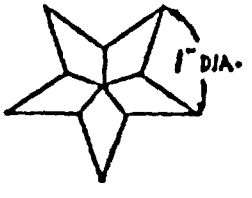
MACHINERY
REPAIRMAN
(MR)



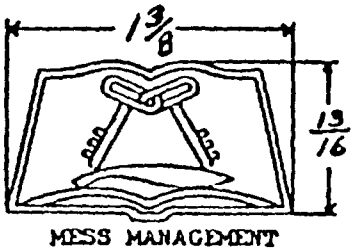
MACHINIST'S
MATE
(MM)



MASTER-AT-ARMS (MA)



Master C.P.O.
of the
Fleet/Command/
FORCE OR NAVY

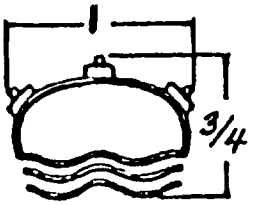


MESS MANAGEMENT
SPECIALIST (MS)

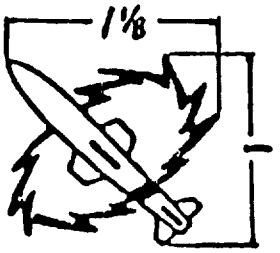
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Figure 9 - Type II - Specialty Mark and Striker Mark

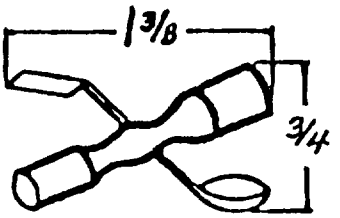
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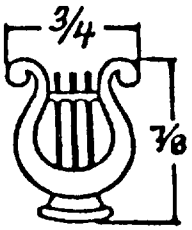
MINEMAN
(MN)



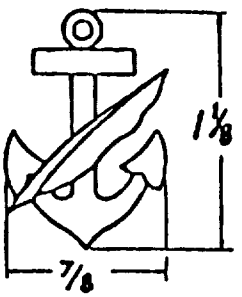
MISSLE
TECHNICIAN
(MT)



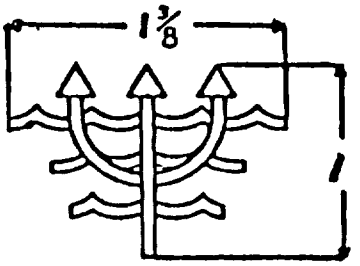
MOLDER
(ML)



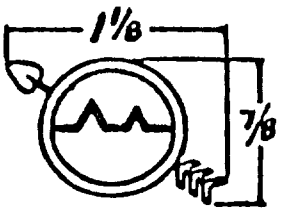
MUSICIAN
(MU)



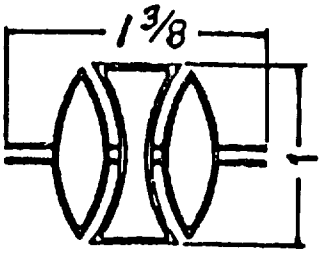
Navy Counselor (NC)



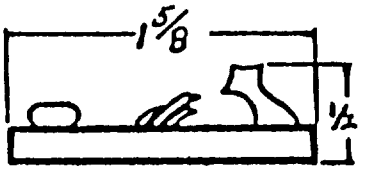
OCEAN SYSTEMS
TECHNICIAN (OT)



OPERATION SPECIALIST
(OS)



OPTICALMAN
(OM)

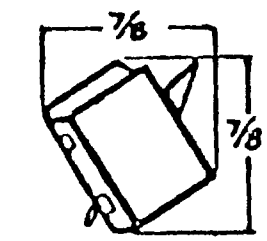


PATTERNMAKER
(PM)

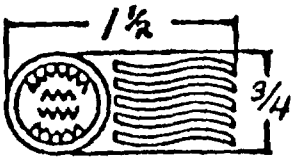
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Figure 10 - Type II - Specialty Mark and Striker Mark

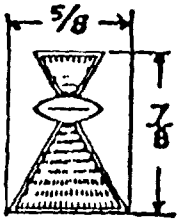
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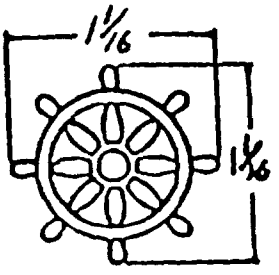
PERSONNELMAN
(PN)



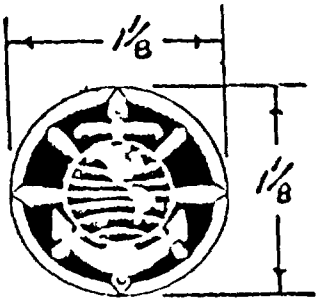
POSTAL CLERK
(PC)



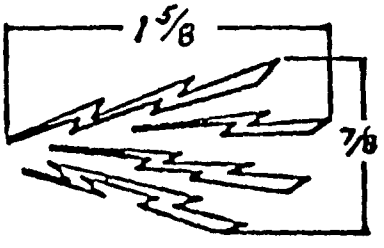
PHOTOGRAPHER'S MATE
(PH)



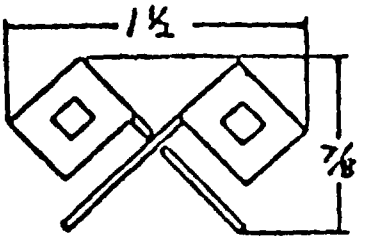
QUARTERMASTER
(QM)



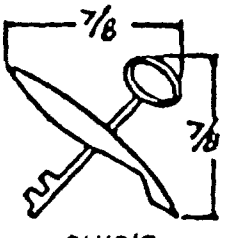
RELIGIOUS PROGRAM SPECIALIST
(RP)



RADIOMAN
(RM)



SIGNALMAN
(SM)

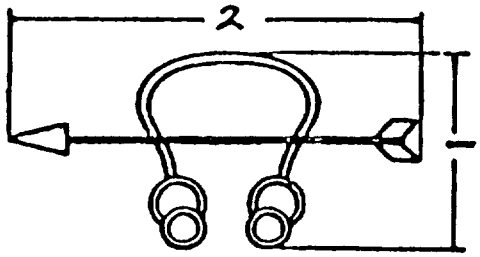


SHIP'S SERVICEMAN
(SH)

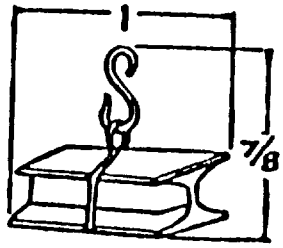
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Figure 11 - Type II - Specialty Mark and Striker Mark

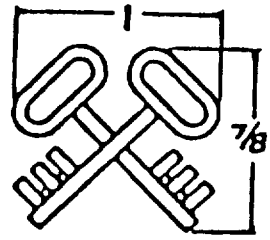
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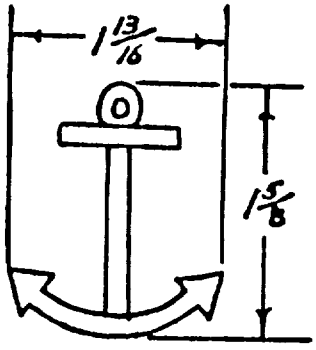
SONAR TECHNICIAN (ST)



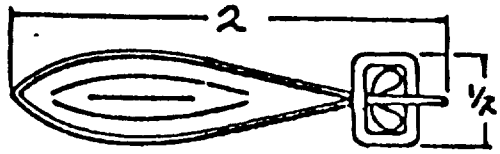
STEELWORKER
(SW)



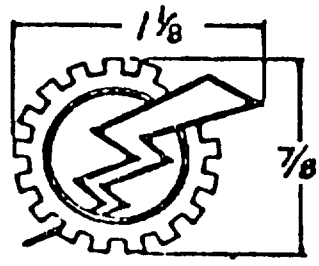
STOREKEEPER
(SK)



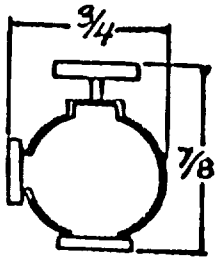
SEAMAN *



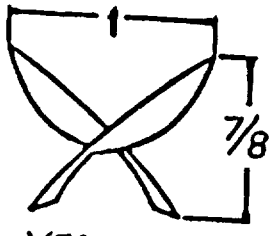
TORPEDOMAN'S MATE
(TM)



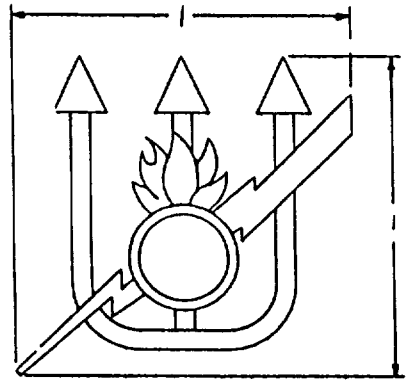
TRADEVMAN
(TD)



UTILITIES MAN
(UT)



YEOMAN
(YN)



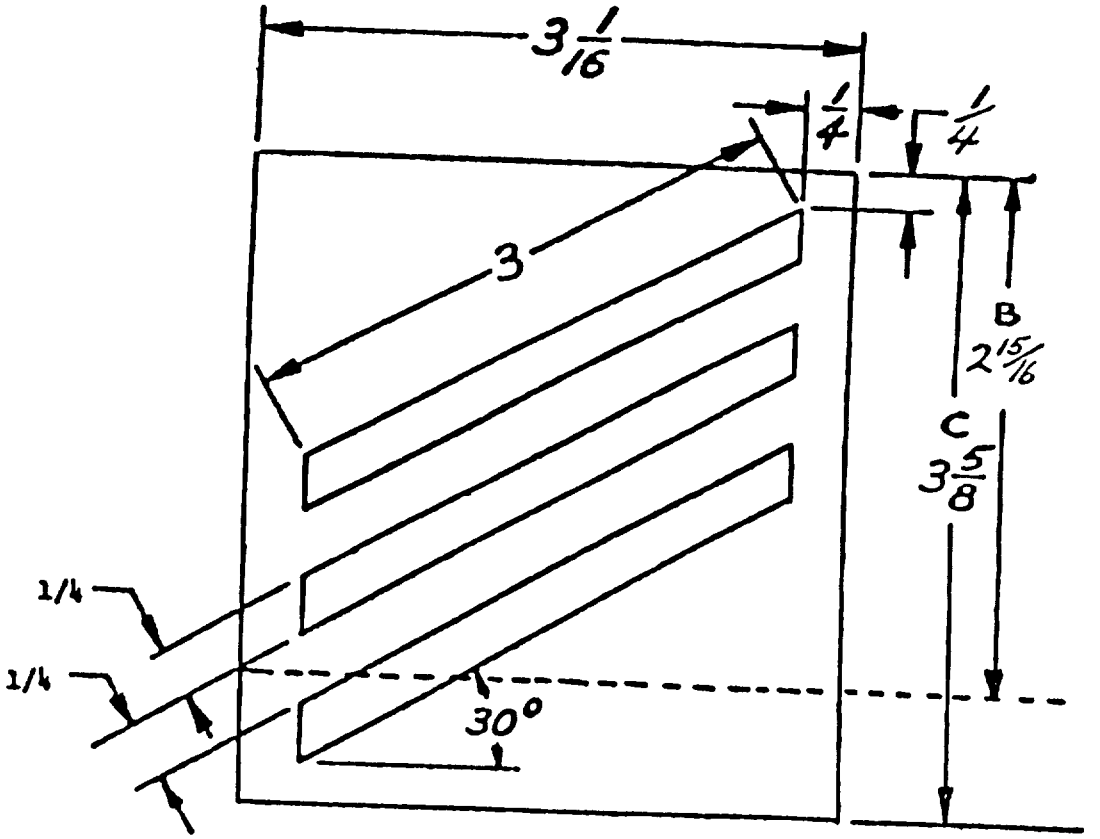
WEAPONS TECHNICIAN (WT)

* Not applicable to Rating Badge

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Figure 12 - Type II - Specialty Mark and Striker Mark

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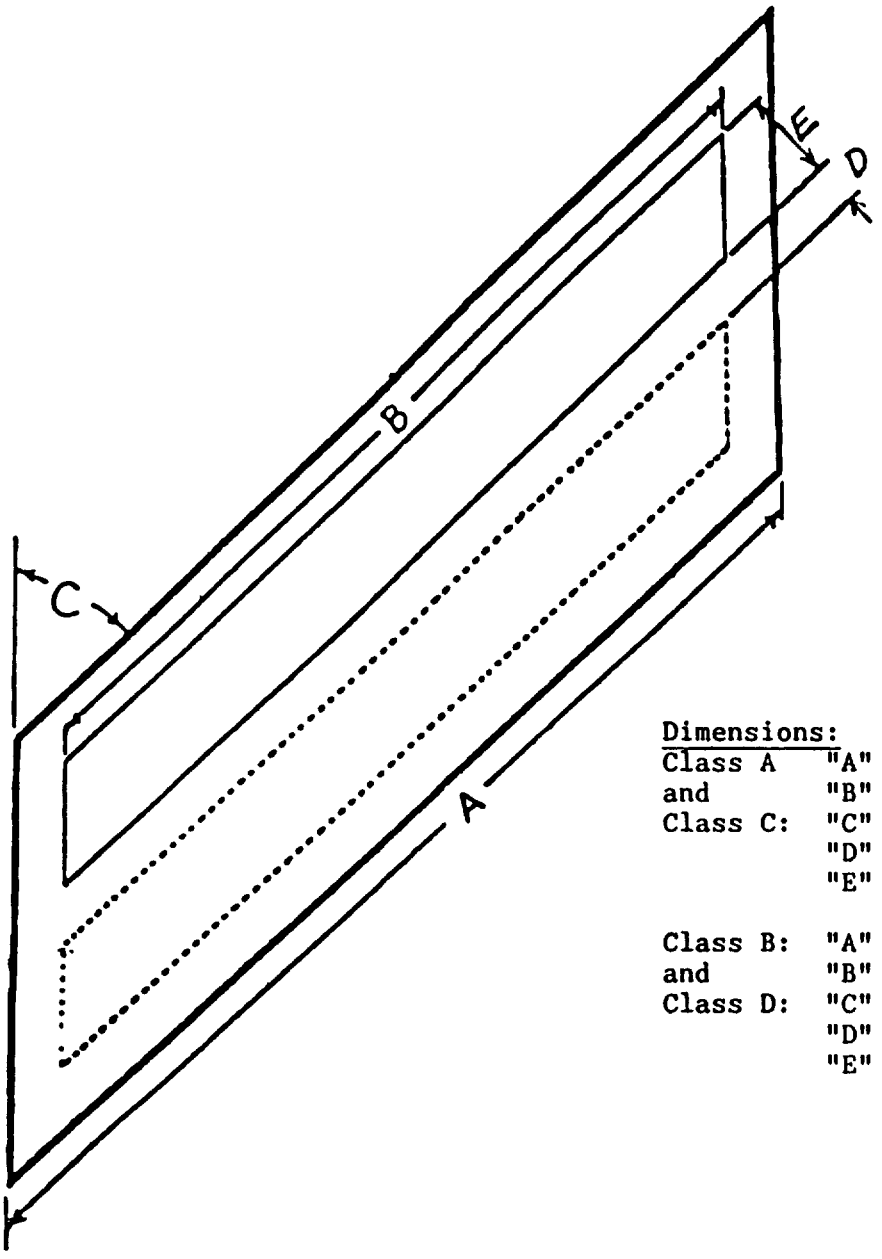


HEIGHT MEASUREMENT
TWO STRIPES = B
THREE STRIPES = C

NOTE: ILLUSTRATION IS FOR DIMENSIONS ONLY. FOR CORRECT DESIGN AND PROPER STITCH EXECUTION REFER TO THE APPLICABLE STITCH ENLARGEMENT. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

Figure 13 - Type V - Group Rate Mark

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Dimensions:

Class A and Class C:	"A" - 7-3/4 inches	
	"B" - 7 inches	
	"C" - 45°	
	"D" - 1/4 inch	
	"E" - 3/8 inch	
Class B and Class D:	"A" - 6 inches	
	"B" - 5-1/4 inches	
	"C" - 45°	
	"D" - 1/4 inch	
	"E" - 3/8 inch	

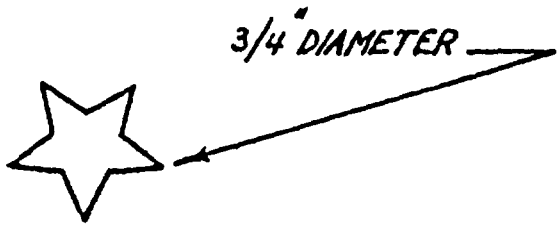
Service Stripe - One or more

BOTH CLASSES SHALL ALLOW A 1/4 INCH BORDER AROUND EMBROIDERED DESIGN

NOTE: ILLUSTRATION IS FOR DIMENSIONS ONLY. FOR CORRECT DESIGN AND PROPER STITCH EXECUTION REFER TO THE APPLICABLE STITCH ENLARGEMENT. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

Figure 14 - Type VI - Stripe, Service

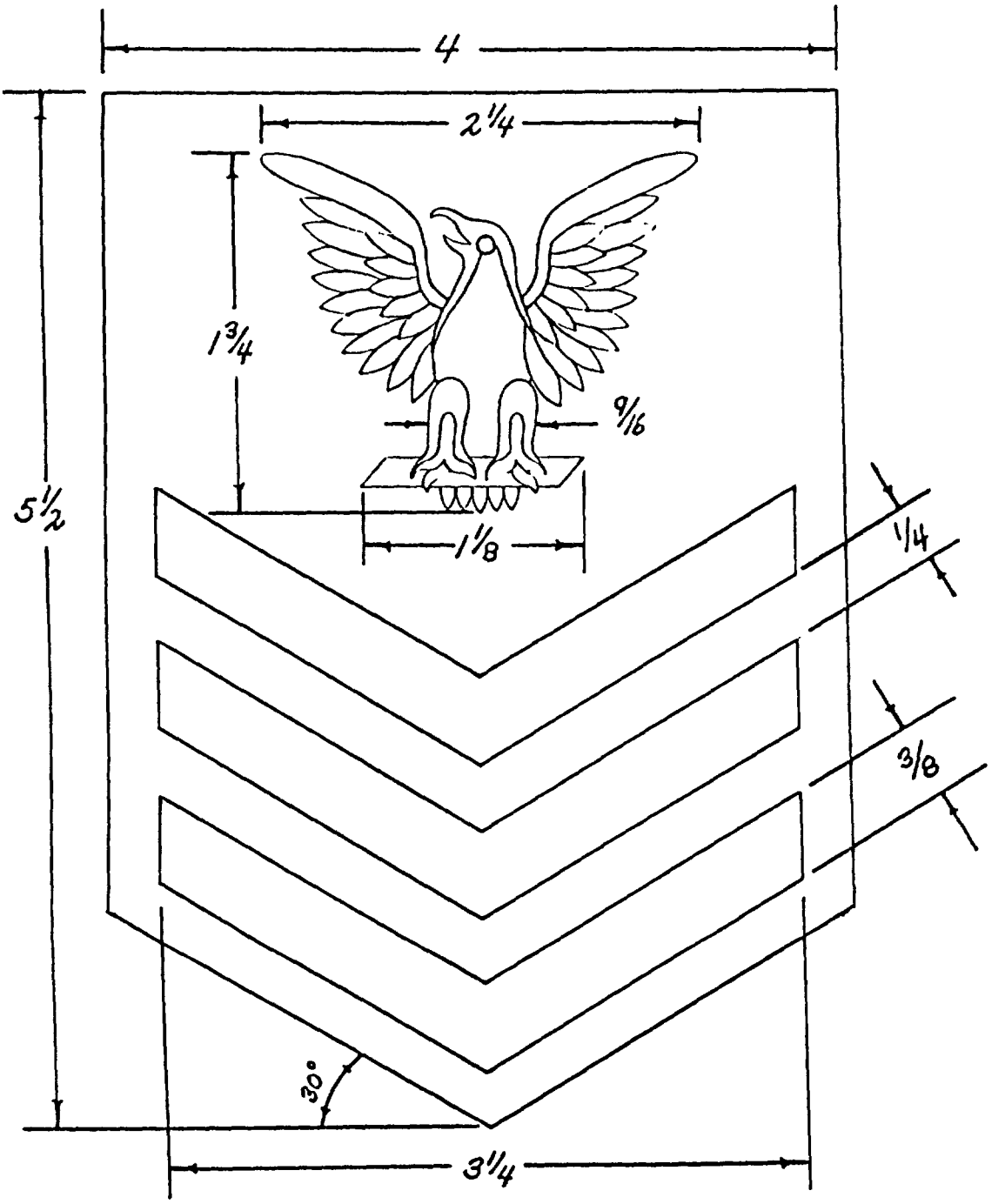
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Figure 15 - Type VIII, Insignia, Grade

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Figure 16 - Type X, Badge, Rating, Utility Petty Officer, First Class

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-I-15864K(NU)	INSIGNIA, EMBROIDERED (MEN AND WOMEN)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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